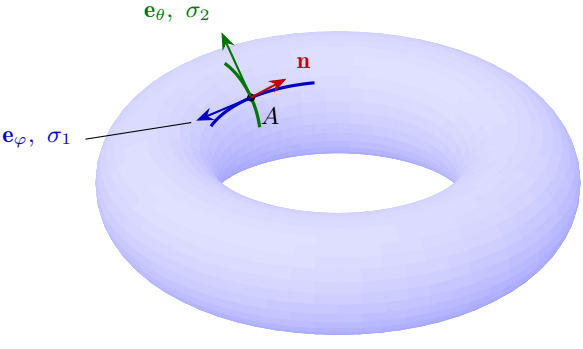
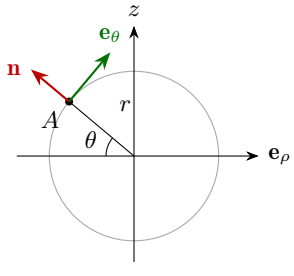


两个力均在膜的切平面内



$$\kappa_\theta = \frac{1}{r}$$

$$\kappa_\varphi = -\frac{\cos\theta}{R - r\cos\theta}$$